Homework 09

The kernel-tuning question moved here from Homework 08. A solution for Question 1 will be available after Monday, October 26, 2026. No solution will be provided for Question 2.

Question 1: Choosing an SVM kernel

Use `saheart.csv` from the combined ZIP. As in Homework 7, predict `chd` using the same nine predictors, coding `famhist` as 1 for `Absent` and 2 for `Present`.

Compare SVMs with a linear kernel, polynomial kernels of degrees 2 (quadratic), 3, and 4, and a radial basis kernel. Choose **no more than 20 candidate settings in total** across these kernels. Vary the cost C for each kernel and the radial scale γ for the radial kernel. State any other kernel settings you hold fixed.

Divide the observations into five folds yourself and use the same folds for all candidates. Standardize predictors using only the fitting observations in each fold. Report the candidate settings and their mean cross-validation misclassification rates. Which kernel and settings do you prefer? Give a brief justification.

Question 1 response

We used the South African heart disease dataset (462 observations) to predict `chd` from all nine predictors, coding `famhist` as 1 for Absent and 2 for Present. We compared linear, polynomial (degrees 2, 3, and 4), and radial basis function (RBF) SVMs. The 20 settings below were evaluated using the same five stratified folds (seed = 432). Predictors were standardized within each fold by fitting the transformation only to the fitting observations. For polynomial kernels, we fixed (=1/9) and `coef0 = 1`; `degree` and `coef0` do not apply to the linear or RBF kernels.

Kernel	Degree	(C)	( )	`coef0`	Mean CV error
Linear	—	0.01	—	—	0.2772
Linear	—	0.1	—	—	0.2706

Kernel	Degree	(C)	( )	`coef0`	Mean CV error
Linear	—	1	—	—	0.2707
Linear	—	10	—	—	0.2750
Polynomial	2	0.1	(1/9)	1	0.2685
Polynomial	2	1	(1/9)	1	0.2814
Polynomial	2	10	(1/9)	1	0.3030
Polynomial	3	0.1	(1/9)	1	**0.2577**
Polynomial	3	1	(1/9)	1	0.3053
Polynomial	3	10	(1/9)	1	0.3375
Polynomial	4	0.1	(1/9)	1	0.2815
Polynomial	4	1	(1/9)	1	0.3180
Polynomial	4	10	(1/9)	1	0.3420
RBF	—	0.1	0.01	—	0.3463
RBF	—	1	0.01	—	0.2728
RBF	—	100	0.01	—	0.2879
RBF	—	0.1	0.1	—	0.3463
RBF	—	1	0.1	—	0.2901
RBF	—	10	0.1	—	0.3094
RBF	—	1	1	—	0.3616

The degree-3 polynomial SVM with (C=0.1), (=1/9), and `coef0 = 1` had the lowest mean cross-validation misclassification rate (0.2577, or 25.77%) among the 20 candidates, so we selected it as our preferred model. Its error was 1.29 percentage points lower than the best linear model (0.2706), 1.08 points lower than the best quadratic model (0.2685), and 1.51 points lower than the best RBF model (0.2728). These are modest differences: the cubic kernel had the lowest error in this comparison, but the results do not establish a substantial or statistically significant predictive advantage.

The analysis script, exact fold assignments, and fold-level errors are saved in q1-svm-cv.py, q1-folds.csv, and q1-cv-results.csv.

Question 2: Write and use a short SVM analysis skill

In Homework 8, you reviewed another student's skill. Review the comments and suggestions you received on your own Homework 7 skill. Use what you learned from both sides of the peer review to write a new, shorter skill for SVM model fitting.

Write the skill by hand. Create a `SKILL.md` with a short name and description in YAML frontmatter. Write the entire file yourself, without asking AI to draft or edit it. The instruction body after the closing `---` of the frontmatter must be **no more than 50 lines**, including blank lines. Keep the skill independent of any particular dataset, variable names, or observed results, and load it only when you explicitly invoke it.

Your skill should direct the agent to:

1. judge whether the data are suitable for an SVM analysis;
2. choose which kernels to consider and explain why;
3. identify tuning parameters and a method for tuning them;
4. choose an evaluation method that supports selecting a final model;
5. require a compact figure and table that justify the choices;
6. interpret the fit and report the key numerical results; and
7. prioritize the evidence needed to justify the final model in **no more than one printed page at 11-point font**.

Apply the skill. Explicitly invoke your skill to analyze `saheart.csv`, using the same response, nine predictors, and `famhist` coding as in Question 1. Provide these data details when invoking the skill, not in the skill itself. Let the skill guide the analysis, then check the resulting report and calculations.

Submit the results. Include the one-page, 11-point analysis report in your Homework 09 submission and append the complete contents of your `SKILL.md` afterward. The appended skill is outside the one-page limit.

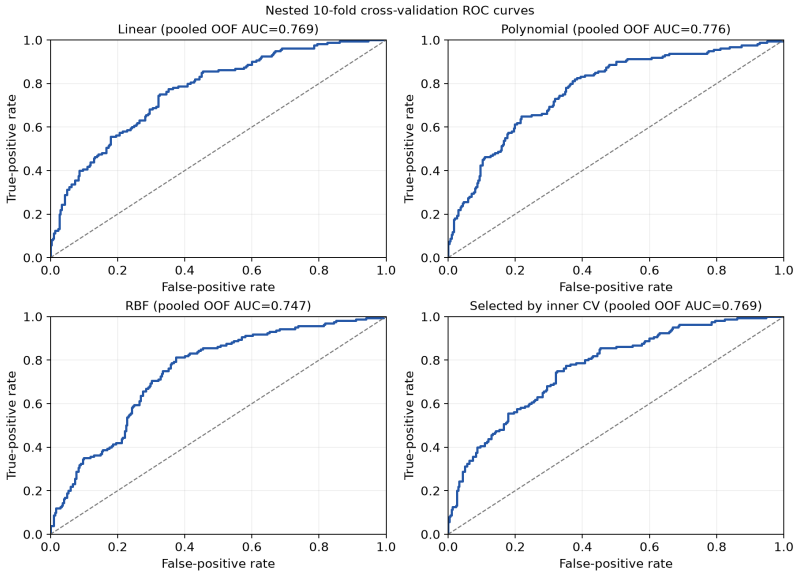
Completed analysis report with the one-page summary and skill appendix: homework-09-q2-report.pdf.

SVM Analysis: South African Heart Disease

Response: CHD (0/1); predictors: all 9 listed variables; family history coded Absent=1 and Present=2. The dataset has 462 observations (302 CHD=0, 160 CHD=1), with no missing values.

Methods: seed 2027; stratified nested 10-fold CV with shared outer folds. Standardization was fitted inside each training fold. Inner 10-fold CV tuned C over 0.01-100 for all kernels; RBF gamma over 0.001-1; polynomial degree over 2-4 and gamma over 0.01-1 (coef0=1). Selection used ROC-AUC and the one-standard-error rule, preferring linear, then polynomial, then RBF when within one SE of the best inner-CV score. Balanced accuracy is also reported.

Kernel	Outer ROC-AUC mean +/- SD	Balanced acc. mean +/- SD	Outer-fold SV mean +/- SD
Linear	0.769 +/- 0.069	0.667 +/- 0.066	253.7 +/- 4.5
Polynomial	0.782 +/- 0.066	0.661 +/- 0.071	269.9 +/- 3.0
Rbf	0.770 +/- 0.067	0.606 +/- 0.115	275.9 +/- 15.0
Selected procedure	0.769 +/- 0.069	0.667 +/- 0.066	See family rows



Final model: Linear SVM (C=0.1, kernel=linear); 283 support vectors (61.3% of N). The outer-CV-selected procedure achieved ROC-AUC 0.769 +/- 0.069 and balanced accuracy 0.667 +/- 0.066. Top held-out permutation-importance features (mean ROC-AUC decrease): age (0.050); famhist (0.036); tobacco (0.028).

Interpretation: the selected kernel is preferred under the stated nested-CV selection rule, not proven universally best. Fold-to-fold SD describes split variation, not a confidence interval. Permutation importance is predictive, not causal; correlated predictors can share or mask importance. No imputation was needed. The full-data refit is for deployment/submission, not an independent test-set estimate.

Reproducibility: seed 2027; scikit-learn 1.9.1; NumPy 2.5.4; pandas 3.1.0; Python 3.12.2. Detailed folds and results: kernel-cv-summary.csv; environment: environment.txt.

Appendix: Author-written SVM skill

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name: svm-analysis
description: Provides clear and correct guidance during the process of building a SVM model.

Please read each instruction carefully, following the order provided below.
1. Input Preparation
   - Specify dataset and its path, the target variable (can be binary or multiclass), and predictor variable list.
   -Define rules for handling missing values, and ensure that preprocessing and standardization are fitted only on the training set of each fold to prevent data leakage.
   - If deafults are missing: set random seed to 2027, output folder to "./results".

2. Kernel Selection Rule
   - Check number of samples in relation to features, class balance, and missing data.
   - Select kernels strictly based on this context: linear kernel for easier interpretation, RBF kernel for better flexibility, and polynomial kernel for catching more lower-order interactions.

3. Parameter Tuning & Cross-Validation
   - Focus tuning parameters on regularization, kernel coefficient, and polynomial degree.
   - Use 10-fold stratified cross-validation, trying to find best parameters on a logarithmic scale.
   - Finally evaluate models strictly based on metrics such as ROC-AUC or another appropriate one; when performance is similar, prefer the simpler model.

4. Deliverables
   - Save files including trained model, cross-validation metrics in a CSV file, and diagnostic ROC grid plot.Then briefly document computational environment
   - Specifically, include a table (metrics $\pm$ SD, # support vectors), a diagnostic plot, the report N_{sv}/N ratio and feature importance.

5. Reporting
   - Present findings on one page using 11-point font, highlighting most significant evidence.(Bold text or any other format you deem appropriate)
   - Clearly separate the core conclusions, which should fit within a single page, from the code and detailed processes, and appendices that follow.